

Chapter 7

Tests of Hypotheses and Significance

Learning Objectives

- Structure engineering decision-making problems as hypothesis tests.
- Test hypotheses on the mean of a normal distribution using either a Z -test or a t -test procedure.
- Test hypotheses on the variance or standard deviation of a normal distribution.
- Test hypotheses on a population proportion.

7.0 Tests of Hypotheses and Significance

7.1 Introduction

We now discuss the subject of hypothesis testing, which as earlier noted is one of the two basic classes of statistical inference. Testing of hypotheses involves using statistical inference to test the validity of postulated values for population parameters. If the hypothesis specifies the distribution completely it is called simple, otherwise it is called composite. For example, a demographer interested in the mean age of residents in a certain local government area might pose a simple hypothesis such as $\mu = 24$ or he might specify a composite hypothesis as $\mu < 24$ or $\mu > 24$.

A statistical test is usually structured in terms of two mutually exclusive hypotheses referred to as the null hypothesis and the alternative hypothesis denoted by H_0 and H_1 respectively.

Two types of error occur in hypothesis testing; these are type I error and type II error. Type I error occurs if H_0 is rejected when it is true. The probability of a type I error is the conditional probability, $P(\text{reject } H_0 | H_0 \text{ is true})$ and is denoted by α .

Hence,

$$\alpha = P(\text{reject } H_0 | H_0 \text{ is true}) \text{ and}$$

$$1 - \alpha = P(\text{accept } H_0 | H_0 \text{ is true})$$

Type II error if H_0 is accepted when it is false. Its probability is denoted by the symbol β , where

$$\beta = P(\text{accept } H_0 | H_0 \text{ is false}) \text{ and}$$

$$1 - \beta = P(\text{reject } H_0 | H_0 \text{ is false}) \text{ called power of the test}$$

Types I and II error can be explained as follows:

	H_0 is true	H_0 is false
Accept H_0	$1 - \alpha$ (correct decision)	β (Type II errors)
Reject H_0	α (Type I errors)	$1 - \beta$ (correct decision)

Standard format of hypothesis testing: this format involves **five** steps.

Step 1: State the null and alternative hypotheses.

Step 2: Determine the suitable test statistics. This involves choosing the appropriate random variable to use in deciding to accept or reject the null hypothesis.

Unknown Parameter	Appropriate Test Statistic
' μ ' σ known, population normal	$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$
' μ ' σ unknown, population normal	$Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$ if n is 'large' usually $n \geq 30$
' μ ' σ unknown, n small, population normal	$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$, with $(n-1)$ df
' p ' population normal, n large	$Z = \frac{(x/n) - p_0}{\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}}$

Step 3: Determine the critical region using the cumulative distribution table for the test statistic. The set of values that lead to the rejection of the null hypothesis is called the critical region. A statistical test may be a one-tail or two-tail test. Whether one uses a one- or two- tail test of significance depends upon how the alternative hypothesis is formulated.

Types of Hypothesis	H_0	H_1	Decision Rule Rejected H_0 if
Two-tail	$\mu = \mu_0$	$\mu \neq \mu_0$	$z > z_{\alpha/2}$ or $z < -z_{\alpha/2}$
Right-tail	$\mu \leq \mu_0$	$\mu > \mu_0$	$z > z_{\alpha/2}$
Left-tail	$\mu \geq \mu_0$	$\mu < \mu_0$	$z < -z_{\alpha/2}$

Step 4: Compute the values of the test statistic based on the sample information, e.g. Z_c , t_c , χ_c^2 .

Step 5: Make a statistical decision and interpretation. H_0 is rejected if the computed value of the test statistic falls in the critical region otherwise it is accepted.

Possible situation in testing a statistical hypothesis

	Hypothesis is correct	Hypothesis is incorrect
Hypothesis is Accepted	Correct decision	Type II error β
Hypothesis is Rejected	Type I error α	Correct decision

Type I error: We reject a hypothesis when it should be accepted. $P(\text{reject } H_0 | H_0 \text{ true})$.

Type II error: We accept a hypothesis when it should be rejected. $P(\text{accepting } H_0 | H_0 \text{ false})$.

7.2 A Single Population Mean,

We shall consider testing of hypothesis about a population mean under three different conditions:

1. When sampling is from a normally distributed population with known variance.

1. When sampling is from a normally distributed population with unknown variance.
2. When sampling is from a population that is not normally distributed.

Sampling From Normally Distributed Populations: Population Variance Known

Example 76

A researcher is interested in the mean level of some enzyme in a certain population. The data available to the researcher are the enzyme determinations made on a sample of 10 individuals from the population of interest, and the sample mean is 22. If the sample came from a population that is normally distributed with a known variance, $\sigma^2 = 45$. Can the researcher conclude that

the mean enzyme level in this population is different from 25? Take $\alpha = 0.05$.

Solution

Step 1: $H_0 : \mu_0 = 25$

$$H_1 : \mu_0 \neq 25$$

Step 2:

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

since μ_0 and σ are known.

Step 3:

Step 4: $\sigma^2 = 45$, $n = 10$, $\bar{x} = 22$

$$Z_c = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

=

$$\frac{22 - 25}{\sqrt{45}/\sqrt{10}}$$

=

$$\frac{-3}{2.1213}$$

=

$$-1.41$$

Step 5: We are unable to reject the null hypothesis, since $-1.42 > -1.96$.

Example 77

Aircrew escape systems are powered by a solid propellant. The burning rate of this propellant is an important product characteristic. Specifications require that the mean burning rate must be 50 centimeters per second. We know that the standard deviation of burning rate is $\sigma = 2$ centimeters per second. The experimenter decides to specify a type I error probability or significance level of $\alpha = 0.05$ and selects a random sample of $n = 25$ and obtains a sample average burning rate of $\bar{x} = 51.3$ centimeters per second. What conclusions should be drawn?

Solution

$$H_0 : \mu_0 = 50$$

$$H_1 : \mu_0 \neq 50$$

$$Z_c = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

=

$$\frac{51.3 - 50}{2/\sqrt{25}}$$

=

$$\frac{1.3}{0.4}$$

= 3.25

Conclusion: Since $z_c = 3.25 > 1.96$, we reject $H_0 : \mu_0 = 50$ at the 0.05 level of significance. We conclude that the mean burning rate differs from 50 centimeters per second, based on a sample of 25 measurements.

7.3 Tests on the Mean of a Normal Distribution: Variance Unknown

Test statistic is

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

has a t distribution with $n-1$ degrees of freedom.

Example 78

A study revealed that the upper limit of the Normal Body Temperature of males is 98.6. The body temperatures for 25 male subjects were taken and recorded as follows: 97.8, 97.2, 97.4, 97.6, 97.8, 97.9, 98.0, 98.0, 98.0, 98.1, 98.2, 98.3, 98.3, 98.4, 98.4, 98.4, 98.5, 98.6, 98.7, 98.8, 98.8, 98.9, 98.9 and 99.0. Test the hypothesis $H_0 : \mu_0 = 98.6$ versus $H_1 : \mu_0 \neq 98.6$, using $\alpha = 0.05$.

Example 79

Nine patients suffering from the same physical handicap, but otherwise comparable were asked to perform a certain task as part of an experiment. The average time required to perform the task was seven minutes with a standard deviation of two minutes. Assuming normality, can we conclude that the true mean time required to perform the task by this type of patient is at least ten minutes?

Example 80

The increased availability of light materials with high strength has revolutionized the design and manufacture of golf clubs, particularly drivers. Clubs with hollow heads and very thin faces can result in much longer tee shots, especially for players of modest skills. This is due partly to the “spring-like effect” that the thin face imparts to the ball. Firing a golf ball at the head of the club and measuring the ratio of the outgoing velocity of the ball to the incoming velocity can quantify this spring-like effect. The ratio of velocities is called the coefficient of restitution of the club. An experiment was performed in which 15 drivers produced by a particular club maker were selected at random and their coefficients of restitution measured. In the experiment, the golf balls were fired from an air cannon so that the incoming velocity and spin rate of the ball could be precisely controlled. Determine if there is evidence (with $\alpha = 0.05$) to support a claim that the mean coefficient of restitution exceeds 0.82. The observations are:

0.8411 0.8191 0.8182 0.8125 0.8750

0.8580 0.8532 0.8483 0.8276 0.7983

0.8042 0.8730 0.8282 0.8359 0.8660

The sample mean and sample standard deviation are $\bar{x} = 0.83725$ and $s = 0.02456$.

7.4 Tests on a Population Proportion

It is often necessary to test hypotheses on a population proportion. For example, suppose that a random sample of size n has been taken from a large population and that $X (\leq n)$ observations in this sample belong to a class having a particular characteristic of interest. Then $\hat{P} = X/n$ is a point estimator of the proportion of the population p that belongs to this class. Note that n and p are the parameters of a binomial distribution. Recall that the sampling distribution of $\hat{P}$ is approximately normal with mean p and variance $p(1-p)/n$, if p is not too close to either 0 or 1 and if n is relatively large.

In many engineering problems, we are concerned with a random variable that follows the binomial distribution. For example, consider a production process that manufactures items that are classi-

fied as either acceptable or defective. It is usually reasonable to model the occurrence of defectives with the binomial distribution, where the binomial parameter p represents the proportion of defective items produced. Consequently, many engineering decision problems include hypothesis testing about p .

Consider testing

$$H_0 : p = p_0$$

$$H_1 : p \neq p_0$$

For large samples, the normal approximation to the binomial with the test statistic

$$Z = \frac{X - np_0}{\sqrt{np_0(1 - p_0)}} = \frac{X/n - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}} = \frac{\bar{P} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

may be used.

This presents the test statistic in terms of the sample proportion instead of the number of items X in the sample that belongs to the class of interest.

Example 81

In a study designed to assess the relationship between a certain drug and a certain anomaly in chick embryos, 50 fertilized eggs were injected with the drug on the fourth day of incubation. On the twentieth day of incubation the embryos were examined and in 12 the presence of the abnormality was observed. Test the null hypothesis that the drug causes abnormalities in not more than 20 percent of eggs into which it is introduced. Let $\alpha = 0.05$.

Example 82

A manufacturer of intraocular lenses is qualifying a new grinding machine and will qualify the machine if the percentage of polished lenses that contain surface defects does not exceed 2%. A random sample of 250 lenses contains six defective lenses. Formulate and test an appropriate set of hypotheses to determine if the machine can be qualified. Use $\alpha = 0.05$.

Example 83

A semiconductor manufacturer produces controllers used in automobile engine applications. The customer requires that the process fallout or fraction defective at a critical manufacturing step not exceed 0.05 and that the manufacturer demonstrate process capability at this level of quality using $\alpha = 0.05$. The semiconductor manufacturer takes a random sample of 200 devices and finds that four of them are defective. Test the null hypothesis that the process fallout does not exceed 0.05.

7.5 The Difference Between two Population Means

Hypothesis testing involving the difference between two population means is most frequently employed to determine whether or not it is reasonable to conclude that the two are unequal. In such cases, one or other of the following hypotheses is tested:

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_1 : \mu_1 - \mu_2 \neq 0$$

$$H_0 : \mu_1 - \mu_2 \geq 0$$

$$H_1 : \mu_1 - \mu_2 < 0$$

$$H_0 : \mu_1 - \mu_2 \leq 0$$

$$H_1 : \mu_1 - \mu_2 > 0$$

Hypothesis Tests for a Difference in Means, Variances Known

Test statistic is

$$Z = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Example 84

In a large hospital for the treatment of the mentally retarded, a sample of 12 individuals with mongolism yielded a mean serum uric acid value of $\bar{x}_1 = 4.5$ mg/100 ml. In a general hospital, a sample of 15 normal individuals of the same age and sex were found to have a mean value of $\bar{x}_2 = 3.4$ mg/100 ml. If it is reasonable to assume that the two populations of values are normally distributed with variances equal to 1. Do these data provide sufficient evidence to indicate a difference in mean serum uric acid levels between normal individuals and individuals with mongolism? Let $\alpha = 0.05$.

Solution

$$n_1 = 12$$

$$n_2 = 15$$

$$\sigma_1^2 = 1$$

$$\sigma_2^2 = 1$$

$$\bar{x}_1 = 4.5$$

$$\bar{x}_2 = 3.4$$

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_1 : \mu_1 - \mu_2 \neq 0$$

$$\begin{aligned}
Z_c &= \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \\
&= \frac{(4.5 - 3.4) - 0}{\sqrt{\frac{1}{12} + \frac{1}{15}}} \\
&= \frac{1.1}{\sqrt{0.15}} = \frac{1}{0.3873} \\
&= 2.84
\end{aligned}$$

Reject H_0 since $2.84 > 1.96$ on the basis of these data, there is an indication that the means are not equal.

Example 85

A product developer is interested in reducing the drying time of a primer paint. Two formulations of the paint are tested; formulation 1 is the standard chemistry, and formulation 2 has a new drying ingredient that should reduce the drying time. From experience, it is known that the standard deviation of drying time is 8 minutes, and this inherent variability should be unaffected by the addition of the new ingredient. Ten specimens are painted with formulation 1, and another 10 specimens are painted with formulation 2; the 20 specimens are painted in random order. The two sample average drying times are $\bar{x}_1 = 121$ minutes and $\bar{x}_2 = 112$ minutes, respectively. What conclusions can the product developer draw about the effectiveness of the new ingredient, using $\alpha = 0.05$?

Solution

$$n_1 = 10$$

$$n_2 = 10$$

$$\sigma_1^2 = 64$$

$$\sigma_2^2 = 64$$

$$\bar{x}_1 = 121$$

$$\bar{x}_2 = 112$$

$$H_0 : \mu_1 \leq \mu_2 = 0$$

$$H_1 : \mu_1 > \mu_2$$

$$\begin{aligned}
Z_c &= \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \\
&= \frac{(121 - 112) - 0}{\sqrt{\frac{64}{10} + \frac{64}{10}}} \\
&= \frac{9}{\sqrt{12.8}} = \frac{9}{3.5777} \\
&= 2.52
\end{aligned}$$

Conclusion: Reject H_0 .

Exercises

i. Two machines are used for filling plastic bottles with a net volume of 16.0 ounces. The fill volume can be assumed normal, with standard deviation $\sigma_1 = 0.020$ and $\sigma_2 = 0.025$ ounces. A member of the quality engineering staff suspects that both machines fill to the same mean net volume, whether or not this volume is 16.0 ounces. A random sample of 10 bottles is taken from the output of each machine.

Machine 1	Machine 2		
16.03	16.01	16.02	16.03
16.04	15.96	15.97	16.04
16.05	15.98	15.96	16.02
16.05	16.02	16.01	16.01
16.02	15.99	15.99	16.00

Do you think the engineer is correct? Use $\alpha = 0.05$.

ii. Two different formulations of an oxygenated motor fuel are being tested to study their road octane numbers. The variance of road octane number for formulation 1 is $\sigma_1^2 = 1.5$, and for formulation 2 it is $\sigma_2^2 = 1.2$. Two random samples of size $n_1 = 15$ and $n_2 = 20$ are tested, and the mean road octane numbers observed are $\bar{x}_1 = 89.6$ and $\bar{x}_2 = 92.5$. Assume normality, and if formulation 2 produces a higher road octane number than formulation 1, the manufacturer would like to detect it. Formulate and test an appropriate hypothesis, using $\alpha = 0.05$.

Hypothesis Tests for a Difference in Means, Variances unknown but Assumed Equal.

We now extend the results of the previous lecture to the difference in means of the two distributions when the variances of both distributions σ_1^2 and σ_2^2 are unknown. If the sample sizes n_1 and n_2 exceed 30, the normal distribution procedures could be used. However, when small samples are taken, we will assume that the populations are normally distributed and base our hypothesis tests on the t distribution. This nicely parallels the case of inference on the mean of a single sample with unknown variance.

The normality assumption is required to develop the test procedure, but moderate departures from normality do not adversely affect the procedure. Two different situations must be treated. In the first case, we assume that the variances of the two normal distributions are unknown but equal; that is, $\sigma_1^2 = \sigma_2^2 = \sigma^2$. In the second, we assume that they are unknown and not necessarily equal.

The test statistic is

$$t_c = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{S_p^2}{n_1} + \frac{S_p^2}{n_2}}} = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

with

$$n_1 + n_2 - 2$$

df

The two sample variances are combined to form an estimator of σ^2 . The **pooled estimator** of σ^2 is defined as follows.

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

Example 86

The diameter of steel rods manufactured on two different extrusion machines is being investigated. Two random samples of sizes $n_1 = 15$ and $n_2 = 17$ are selected, and the sample means and sample variances are $\bar{x}_1 = 8.73$, $s_1^2 = 0.35$, $\bar{x}_2 = 8.68$, and $s_2^2 = 0.40$, respectively. Assume that $\sigma_1^2 = \sigma_2^2$ and that the data are drawn from a normal distribution. Is there evidence to support the claim that the two machines produce rods with different mean diameters? Use $\alpha = 0.05$ in arriving at this conclusion.

Example 87

Two catalysts are being analyzed to determine how they affect the mean yield of a chemical process. Specifically, catalyst 1 is currently in use, but catalyst 2 is acceptable. Since catalyst 2 is cheaper, it should be adopted, providing it does not change the process yield. A test is run in the pilot plant and results in the data shown in the following table. Is there any difference between the mean yields? Use $\alpha = 0.05$, and assume equal variances.

Observation Number	Catalyst 1	Catalyst 2
1	91.50	89.19
2	94.18	90.95
3	92.18	90.46
4	95.39	93.21
5	91.79	97.19
6	89.07	97.04
7	94.72	91.07
8	89.21	92.75
	$\bar{x}_1 = 92.255$	$\bar{x}_2 = 92.733$
	$s_1 = 2.39$	$s_2 = 2.98$

Example 88

Serum amylase determinations were made on a sample of 15 apparently normal subjects. The sample yielded a mean of 96 units/100ml and a standard deviation 35 units/100ml. Serum amylase determinations were also made on 22 hospitalized subjects. The mean and standard deviation from this second group are 120 and 40 units/100ml., respectively. Would we be justified in calculating that the implied population means are different? Let $\alpha = 0.05$.

Hypothesis Tests for a Difference Between Two Population Proportions

Suppose that two independent random samples of sizes n_1 and n_2 are taken from two populations, and let X_1 and X_2 represent the number of observations that belong to the class of interest in

samples 1 and 2, respectively. Furthermore, suppose that the normal approximation to the binomial is applied to each population, so the estimators of the population proportions $\hat{P}_1 = X_1/n_1$ and $\hat{P}_2 = X_2/n_2$ have approximate normal distributions.

The test statistic is

$$Z = \frac{\bar{P}_1 - \bar{P}_2 - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}}$$

is distributed approximately as standard normal and is the basis of a test for $H_0 : p_1 = p_2$. If the null hypothesis is true, using the fact that $p_1 = p_2 = p$, the random variable

$$Z = \frac{\hat{P}_1 - \hat{P}_2}{\sqrt{p(1-p) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

is distributed approximately $N(0, 1)$. An estimator of the common parameter p is

$$\hat{P} = \frac{X_1 + X_2}{n_1 + n_2}$$

The **test statistic** for $H_0 : p_1 = p_2$ is then

$$Z = \frac{\hat{P}_1 - \hat{P}_2}{\sqrt{\hat{p}(1-\hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

Example 89

A random sample of 500 adult residents of Maricopa County found that 385 were in favor of increasing the highway speed limit to 75 mph, while another sample of 400 adult residents of Pima County found that 267 were in favor of the increased speed limit. Do these data indicate that there is a difference in the support for increasing the speed limit between the residents of the two counties? Use $\alpha = 0.05$.

Example 90

Out of a sample of 150, selected from patients admitted over a two-year period to a large hospital, 129 had some type of hospitalization insurance. In a sample of 160 similarly selected patients from a second hospital, 144 had some type of hospitalization insurance. Test the null hypothesis that $p_1 = p_2$. Let $\alpha = 0.05$.